

Strengthening & Toughening

Previously, we discussed how atomic bonding of different materials affect their macroscopic properties. However, material properties can be further manipulated by changing the microstructure of the material, by mixing elements, heating, and mechanical methods.

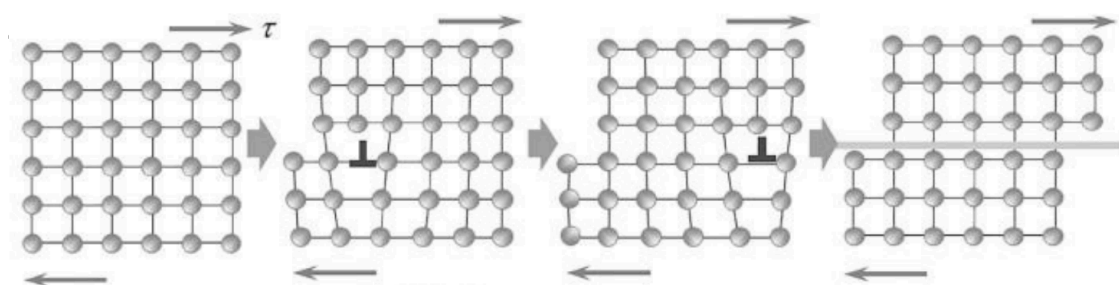
Dislocations in Crystalline Materials

Firstly, we need to better understand how materials fracture. Theoretically, the (shear) stress required to fracture a material is the force needed to break all the atomic bonds at the same time (that are contained in one unit area).

However, the recorded stress tends to be 1000 to 10000 times lower, due to microscopic defects.

Typically for metals, one reason why this value is so low is because actual shearing does not involve simultaneously breaking all the atomic bonds, as the force and energy needed would exceed the strength of the crystals.

Instead, shearing involves multiple steps of dislocations, known as slip. These dislocations are irregularities in the crystal, instead of the atoms being bonded in their regular structure.



Under shear, the dislocations move along a line in the crystal structure, having a total effect of moving a set of atoms across. Because this dislocation is local but moves across the crystal, all the bonds are not broken simultaneously and thus require less energy to shear, making it closer to the actual force needed.

Strengthening Metals

If dislocations were not present in metals, shear forces would need to break every bond simultaneously, reaching theoretical strength. However, this prevents any plastic deformation, as any shear will break all bonds, causing it to be very brittle.

Although dislocations reduce the strength of the metal, their motion allows for plastic deformation and ductility. Hence, the yield strength is related to how much stress is required to move the dislocations. Likewise, if movement of the dislocations is restricted, the metal becomes stronger and harder.

Thus, common methods to strengthen and harden metals involve restricting the motion of dislocations at an atomic scale.

Grain Size Strengthening

Recall that metals have grains, or can also be said to be polycrystalline, as there are multiple grains each with a crystalline structure, which may have dislocations in them. The grain boundaries prevent these dislocations from moving, thus if the metal has finer grains, there are more grain boundaries that restrict the dislocations from moving.

The yield strength can be determined by the Hall-Petch Equation,

$$\sigma_y = \sigma_0 + \frac{k_y}{\sqrt{d}}$$

where σ_0 is the friction stress, k_y is the strengthening coefficient, dependent on material, and d is the average grain diameter.

Solid Solution Strengthening

When alloying elements are added to a metal, they are differently sized than the host metal, distorting the regular crystalline structure, restricting the movement of dislocations and needing higher stress to move them, causing a higher yield strength.

Precipitation Hardening

Alternatively, when alloying elements are added, they react with the host metal to form precipitates, much larger than single alloying atoms (or solute atoms) in the metal, causing distortions in the crystalline structure. Similarly, precipitates restrict dislocations from moving, requiring more stress to overcome the distortions, increasing yield strength.

Work Hardening

Work hardening involves plastically deforming the metal through forging, bending, or other mechanical operations. These processes move dislocations through the metal as it is deformed, which create many more dislocations. When these dislocations interact, they tangle and block one another, further restricting their movement.

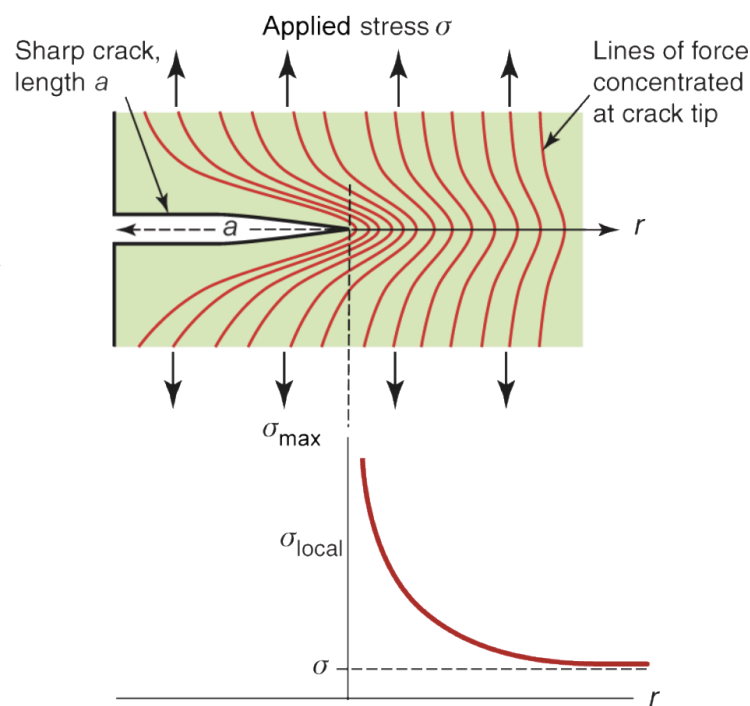
Thus, more stress is required to overcome the tangles, which increase the yield strength of the metal as plastic deformation continues.

The selection of which process to strengthen the metal depends on the use case and performance needed. Typically, as yield strength increases, plastic deformation is more difficult and processes that rely on plastic deformation like ductility and toughness decrease.

Ceramics & Brittle Fracture

Although the crystalline structure of ceramics do also contain dislocations, they are unable to move about because of the electrostatic repulsion of the ions and the directional covalent bonds. However, ceramics still fail below their theoretical strengths because of microscopic imperfections, that have micro-cracks which concentrate stresses.

Around a crack tip, the stress lines concentrate around there, which cumulate together, making the stress around the tip much higher than the applied stress.



The moment the local stress surpasses the theoretical stress, the ceramic bonds fail and the crack quickly propagates until failure. These cracks grow in an unstable manner, self propagating with little energy dissipation.

To prevent this, we can toughen ceramics so that energy is dissipated into mechanical distortion and not into fracturing. Recall metals are able to yield due to their high toughness, making it less likely to fracture. Making ceramics able to absorb energy helps to prevent cracks from propagating, preventing their fracture.

Transformation Toughening

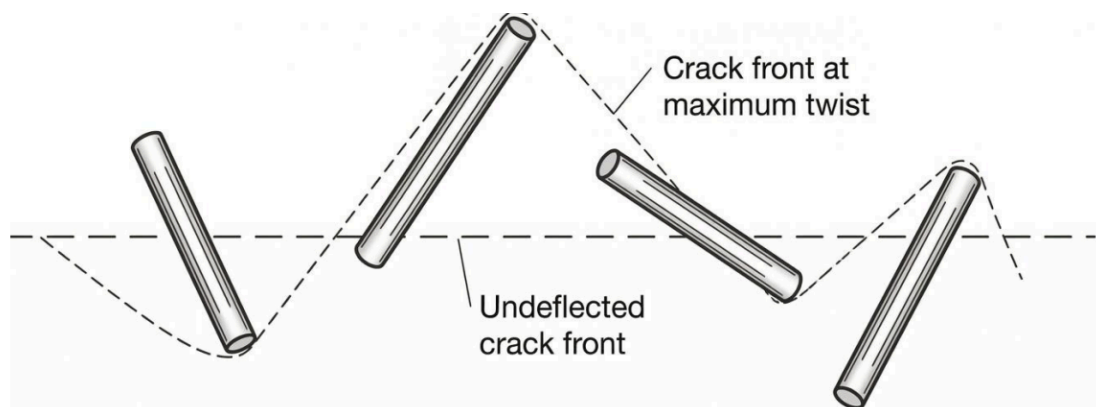
Some ceramics contain different crystal structures that have different volumes than the main crystalline structure of the ceramic. These tend to transform suddenly under high local stress, expanding and absorbing energy.

The expansion of the crystals also compressed the crack, reducing the stress intensity and also requiring more energy to grow the crack (to overcome the expansion)

Crack Deflection

Increasing the length the crack has to propagate along requires more energy, which absorbs more energy and makes the ceramic tougher. Alternatively, deflecting the original direction of the crack also decreases the stress intensity relative to the applied stress.

One method for alumina (Al_2O_3) is by adding SiC fibers, to force the crack to twist and travel along longer paths.



Crack Bridging & Fiber Pull-out

More specifically for fiber composites with a ceramic matrix, the long continuous fibers carry some of the load as the crack tried to propagate under tensile stress.

As cracks form and propagate to reach a fiber interface, the fibers carry some of the load, opposing the crack opening, reducing the stress intensity at the tip. Further load also causes the fibers to fracture instead as they debond, absorbing energy away from the crack propagation, dissipating energy and preventing sudden failure.

